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Machine Learning

Lab 1

Regression Models

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Compare Model Performance (MAE, RMSE and R^2)

MAE stands for Mean Absolute Error. It represents your average predication error. RMSE stands for Root Mean Squared Error and it highlights large errors. For example, if the difference between MAE and RMSE is huge, it means the model occasionally makes larger mistakes since it has outliers it includes. R^2 shows the percentage of price variation explained in our model. For example, $R^2 = 0.96$ means 96%, which is a very strong model. R^2 has the smallest range of errors possible.

When comparing prediction accuracy on the test set, Random Forest achieved lower MAE and RMSE values (2750.561134 and 4254.754939) compared to Linear Regression (4881.631980 and 7039.359927). Additionally, Random Forest gave a higher R^2 score (0.964772 and 0.903570). In practical terms, this indicated that Random Forest makes significantly smaller pricing errors on average and accounts for a much higher proportion of flight price variance.

Influential features

To determine features importance, we extracted the feature weights/tree importance from our models. The analysis revealed that Travel Class (Business vs Economy) was the single most influential feature affecting price, followed by Flight Duration and Number of Stops. Categorical features such as specific airline dummy variables had a secondary impact, showing airline dummy variables had a secondary impact, showing that tickets class and route length drive baseline prices far more than the specific carrier.

As shown in our boxplot analysis, cabin class creates a distinct price split. Business class ticket operates in a completely higher price bracket compared to Economy tickets, which remain mostly under 10.000. This confirms why travel class emerged as the most critical feature in predicting ticket prices.

Our scatterplot of flight duration against price shows two distinct horizontal bands representing Economy and Business. While longer flight durations generally increase ticket price within the same cabin class, the travel class itself sets for foundational baseline.

Comparing Strength and Weakness in the models

Linear regression is easy to understand and explain since the model gives you clear numbers showing exactly how much each feature adds to or subtracts from the ticket price. It also looks at features one by one and it assumes every feature affects the price on its own, missing how features combine (for example, it doesn't automatically realize that long flight in Business class jump in price much faster than long flights in Economy).

Random Forest handles complex combinations well. It easily shows up on how features work together, like knowing that a long flight in Business class on a weekend cost far more than just adding those features separately. It also handles weird or extreme data better. It won't mess up its overall accuracy as easily, leading to fewer precision errors overall.

If you let the decision trees grow too deep without limits, the model might memorize the exact training set instead of learning general rules, making it perform worse on new data. It also takes more computer power to run, and because it uses hundreds of decision trees behind the scenes, it's tricky to see exactly how it reaches a specific price guess.

Recommended Model

Based on our results, we recommend using the Random Forest Regressor to predict flight prices. Random Forest consistently outperformed Linear Regression across all evaluation metrics, delivering a higher R^2 score and lower MAE and RMSE errors. Because flight pricing is inherently non-linear and heavily

influenced by feature interactions (such as travel class combined with flight duration), the tree-ensemble approach provides significantly more reliable pricing estimates for unseen test data.